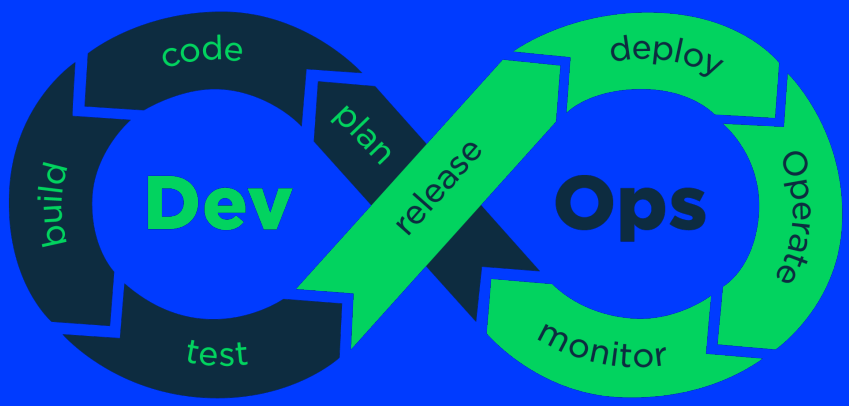
{EPITECH}

CHOCOLATINE_

< IMPROVE INTEGRATION AND TESTING WITH GITHUB ACTIONS />



CHOCOLATINE

Whatever your opinions might be, this is a *chocolatine*.

But what purpose does it serve? It allows you to go through the day and through your work better. In other words, it facilitate your work and allows you to focus on things that matter (discussing its name does not pertain to this category).

Your repositories are hosted on GitHub, and GitHub has an intuitive way of automating the launch of pre-defined actions, triggered by events of your choice. It is, in the same way as chocolatines, a great way to facilitate and enhance your development and your workflow.

In this project, you will setup a GitHub Actions workflow to enforce good practices and diverse rules in a repository.

GitHub Actions are free and unlimited for public repositories, and free up to 2,000 minutes per month for personnal private repositories, which is quite generous!

If particular settings or elements are not specified or addressed in the subject, you are free to do as you please with them.

Technical details

You will have to turn in a single YAML workflow file named `chocolatine.yml`, placed either:

- ✓ at the root of your repository;
- ✓ or in the `.github/workflows` folder.

Your workflow must be usable with your different Epitech projects and their respective technologies and associated tools.

> ⚠ For this project, the **only** external actions allowed are `actions/checkout` and `pixta-dev/repository-mirroring-action`.
> All other external actions (which you can find on the GitHub marketplace for example) are strictly **forbidden**.

Evaluation

Your workflow will be tested by copying your `chocolatine.yml` file, **and only this file**, into a test repository's `.github/workflows` directory.

The environment variables defined later in the subject will be modified to match the test environment; **no other modification will be made to the workflow file**.

Make then sure that the workflow is **self-contained** and does **not need any external files**.

Secrets

You might need to use secured data or values to make your workflow successfully run.

In this case, you **must** use secrets.

> ⚠ **If any hardcoded sensitive value is found in your workflow file, your entire project will be invalidated. You have been warned.**

Features

Using a GitHub Actions workflow, you will need to implement a set of features.

The workflow must be run on every **push** and on every **pull request** creation, **unless**:

- ✓ the branch name starts with "`no-ga/`";
- ✓ or the current repository is the same as the mirror repository.

In any of these two cases, none of the jobs required by the subject must be run at all.

The "not being in the mirror repository" requirement is important, as it will prevent the mirroring from being triggered in the target repository itself.

Triggering an error by exiting with a positive value is **not considered proper handling** of this case.

Have a look at the default environment variables available in the GitHub Actions environment, you might find something useful. ;)

Furthermore, **each job must**:

- ✓ start by **checking out the repository** to the relevant branch;
- ✓ only be run if the **previous job has succeeded**.

The order in which the jobs are expected to be run is the order in which they are defined below in the subject.

Environment variables

You need to define several environment variables, available at the **workflow level**.

- ✓ `MIRROR_SSH_URL`: the **SSH** URL of the Epitech repository which will act as a mirror;
- ✓ `EXECUTABLES`: a comma-separated list of the paths of the executables expected to be produced (it can have a single executable, like `"mysh"` or `"directory1/subdirectory2/my_hunter"`, or several, like `"directory1/executableA,directory2/executableB"`).

All environment variables' values will be strings.

”available at the workflow level” means that the environment variables must be only defined **once**, and be available for **all jobs and steps** of the workflow.

Jobs

Checking the compilation and coding style

Create a job with a `check_compilation_and_coding_style` ID that runs in an `epitechcontent/epitest-docker` Docker container, and that **in order**:

- ✓ launches `banana-check-repo` at the root of the repository;
- ✓ launches `make` at the root of the repository (this **step** must have a 2 minutes timeout);
- ✓ if any, displays each coding style error as an *error annotation* (see next page);
- ✓ makes the job fail if there are any coding style errors;
- ✓ launches `make clean` at the root of the repository;
- ✓ verifies that each file specified in the `EXECUTABLES` environment variable exists and is executable (the job must fail if any of the executables is not there, or not executable).

Each item of the above list must be done on its own **separate** step.

Projects are allowed to use either `epiclang` or the compiler specified in the `CC` environment variable as their compiler in their Makefile.

However, if `$(CC)` is used in the Makefile, without any specific configuration, `clang` (and not `epiclang`) would be used by default, and thus no coding style check would be done, as the Banana plug-in would not be loaded automatically.

Therefore, you **must** ensure that the `CC` environment variable is set to `epiclang` when running `make`.

Editing the Makefile is **not** a correct solution.

If there are any coding style errors, the output of the job must be similar to this:

```
Minor coding style error: main.c#L3
C-L4 - function body opening brace not alone on its line

Minor coding style error: main.c#L3
C-L4 - function body opening brace on same line as prototype

Major coding style error: main.c#L3
C-F6 - function without parameter

Info coding style error: main.c#L6
C-A3 - file not ending with a newline

Minor coding style error: main.c#L1
C-G1 - file not starting with standard Epitech header

Minor coding style error: main.c#L5
C-L2 - indentation not done with groups of 4 spaces

Minor coding style error: main.c#L4
C-L2 - indentation not done with groups of 4 spaces

Minor coding style error: ./src/UglyFile.c#L0
C-O4 - invalid file name

Major coding style error: ./Dockerfile~#L0
C-O1 - unwanted file

Major coding style error: ./#temp_file.sh##L0
C-O1 - unwanted file
```

By using the appropriate error annotation parameters, you **must** be able to click on it and be taken to the precise line of the offending file.

This will only work if the offending line has been introduced or modified in the current commit.

Running tests (because that's what heroes do)

Create a job with a `run_tests` ID that launches `make tests_run` at the root of the repository (this **step** must have a 2 minutes timeout).

This job must must be run in an `epitechcontent/epitest-docker` Docker container.

Pushing to the mirror repository

Create a job with a `push_to_mirror` ID that runs a mirroring to the repository specified in the `MIRROR_SSH_URL` environment variable.

It must use a secret named `GIT_SSH_PRIVATE_KEY` to specify the SSH private key to use.

This job must only be executed when a push is made to the repository.

Good practices

You can do things in lots of different ways with GitHub Actions, but some are better than others.

Here are some examples of good practices to follow:

- ✓ give **meaningful names** to your jobs and steps;
- ✓ use as much **GitHub Actions features** as possible before falling back to custom shell commands;
- ✓ use the **latest** actions' versions;
- ✓ when checking out, only fetch what is **necessary**;
- ✓ display missing or non-executable executables as **error annotations**.

This will be evaluated during the defense.

Bonus

Going further is amazing! Here are some ideas to improve your workflow:

- ✓ check for **banned functions**;
- ✓ add a `LIBRARIES` environment variable to check for the presence of libraries after compilation (useful for projects where you have to make libraries and no executables);
- ✓ make the `check_compilation_and_coding_style` job fail if a compiler other than `epiclang` or the compiler specified in the `CC` environment variable is used.

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